IAT Test 1

Question paper, answer key
and worked solutions

> **MANUAL REVIEW DRAFT**
> 2027 preparation using the verified 2026 pattern and stored 2026 syllabus.
> The 2027 brochure has not been verified. Not approved for student publication.

60	180	240
single-correct questions	minutes	maximum marks

Subject	Question numbers	Marks
Biology	1-15	60
Chemistry	16-30	60
Mathematics	31-45	60
Physics	46-60	60

Instructions

1. There are 60 single-correct questions: Biology 1–15, Chemistry 16–30, Mathematics 31–45 and Physics 46–60.
2. Each question has exactly one correct option. Correct: +4; incorrect: -1; unanswered: 0. All questions are available to attempt; there is no best-three-subject scoring rule.
3. Use 180 minutes. An on-screen non-programmable scientific calculator is part of the 2026 examination conditions; personal calculators are not allowed under that scheme.
4. Treat stated data and idealisations as exact unless an approximation is explicitly supplied. Log denotes base 10 and ln denotes the natural logarithm.
5. The review PDF contains the question paper first, followed by a separate answer key and worked solutions. Keep the solutions hidden during a timed attempt.

Fixed authored order. Figures and IDs must remain linked during review.
Revised 30 October 2026 · IAT27_TEST_01 · Original AI-assisted content.

QUESTION PAPER / B

Biology

Questions 1-15 · 15 questions · 60 marks

Q1

+4 / -1

In plant X, the conspicuous green generation produces gametes; the spore-producing generation remains attached to it and nutritionally dependent on it. In plant Y, the conspicuous green generation produces spores and is independent. Which identification is consistent with these observations?

A) X is a fern and Y is a moss

B) X is a moss and Y is a fern

C) Both X and Y must be mosses

D) Both X and Y must be ferns

Q2

+4 / -1

Two young stem sections are examined. Section P has many scattered, conjoint vascular bundles without cambium between xylem and phloem. Section Q has conjoint bundles in a ring, with cambium between xylem and phloem. Under the usual NCERT model of stem anatomy, which prediction is best supported?

A) P should undergo ordinary cambium-driven secondary growth, but Q should not

B) Both sections are roots because xylem and phloem occur in the same bundle

C) Both sections have closed vascular bundles

D) Q can establish a vascular cambium ring for secondary growth; P normally cannot

Q3

+4 / -1

Three equal preparations of an enzyme are assayed at the same final pH and temperature. U is untreated. I contains a reversible inhibitor. H was heated strongly and then cooled.
Approximate initial rates are:
Low substrate: U = 40, I = 20, H = 4 units.
Very high substrate: U = 100, I = 100, H = 10 units.
Rates are rounded to the experimental precision. Which interpretation is consistent with both substrate conditions?

A) I is consistent with competitive inhibition; heating has reduced the amount of functional enzyme in H

B) I must have destroyed half the enzyme molecules irreversibly; H is fully restored by cooling

C) I has doubled the maximum catalytic rate; H has increased substrate affinity

D) Both I and H merely reduce substrate concentration, with no effect on enzyme function

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Q4

+4 / -1

A test-cross of a double heterozygote produces 420 Ab, 420 aB, 80 AB and 80 ab progeny. Assume these frequencies accurately represent gamete frequencies, recombination is the same in male and female meiosis, and there is no viability selection. If two individuals with this same double-heterozygous genotype and phase are crossed, what percentage of their offspring is expected to be aabb?

A) 6.4%

B) 17.64%

C) 0.64%

D) 25%

Q5

+4 / -1

Urine measurements before and after an intervention are shown. Glomerular filtration and the medullary osmotic gradient stay constant.

Condition	Urine volume (L/day)	Osmolarity (mOsm/L)
Before	2.0	300
After	0.75	800

Which interpretation best combines the measured solute output with the likely tubular response?

A) Solute output falls; collecting-duct water permeability decreases

B) Solute output is unchanged; collecting-duct water permeability increases

C) Solute output increases; collecting-duct water permeability increases

D) Solute output is unchanged; collecting-duct water permeability decreases

Q6

+4 / -1

An ecosystem's plants fix 2400 kJ m−2 day−1 and use 900 kJ m−2 day−1 in respiration. Herbivores assimilate 180 kJ m−2 day−1 of plant-derived energy and respire 120 kJ m−2 day−1; the remainder becomes new herbivore biomass. What percentage of plant net primary production becomes this new herbivore biomass?

A) 4%

B) 7.5%

C) 12%

D) 2.5%

Q7

+4 / -1

A diploid cell has 12 chromosomes in G1. It completes S phase and reaches G2 without mitosis. If its G1 DNA content is 2C, what are its chromosome number and DNA content in G2? Count chromosomes by centromeres, before sister chromatids separate.

A) 24 chromosomes and 4C DNA

B) 12 chromosomes and 2C DNA

C) 24 chromosomes and 2C DNA

D) 12 chromosomes and 4C DNA

Q8

+4 / -1

A diploid female angiosperm has 2n = 24 chromosomes and a tetraploid male of the same species has 4n = 48. Assume normal meiosis, a typical embryo sac with two reduced polar nuclei, successful double fertilisation and no chromosome loss. What chromosome counts and maternal:paternal chromosome contributions are expected in the zygote and primary endosperm nucleus, respectively?

A) Zygote 24, 1:1; endosperm 36, 2:1

B) Zygote 36, 1:2; endosperm 72, 2:1

C) Zygote 36, 1:2; endosperm 48, 1:1

D) Zygote 48, 1:1; endosperm 36, 1:2

Q9

+4 / -1

A sealed thylakoid preparation is placed in darkness with the imposed pH difference shown. Its membrane is otherwise proton-tight, its ATP synthase is functional, and ADP and inorganic phosphate are available outside. No other energy source is supplied. Which short-term observation is expected?

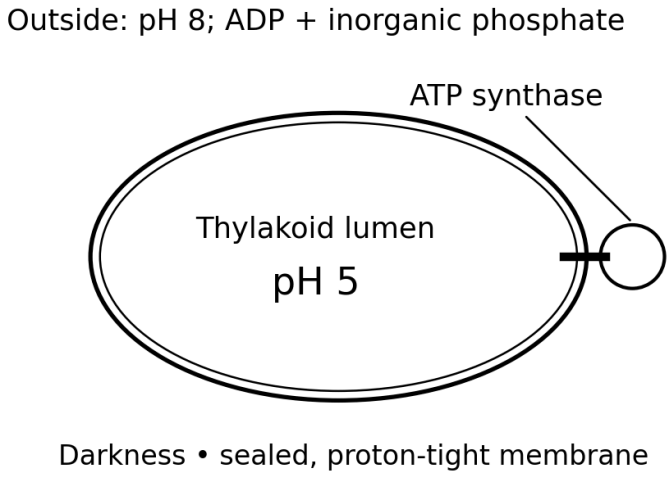


Figure for Q9

A) Transient ATP synthesis; no light-driven NADPH formation

B) Transient NADPH formation; no ATP synthesis from the gradient

C) Neither ATP nor NADPH formation before illumination

D) Both products form while protons enter the acidic lumen

Q10

+4 / -1

In an immunology comparison, preparation P contains ready-made antitoxin antibodies, whereas preparation Q contains a harmless toxoid that elicits the recipient's own immune response. Which comparison is expected in an immunologically responsive recipient?

A) P and Q both supply memory cells and act immediately

B) P supplies prompt protection; Q can induce delayed immune memory

C) Q supplies prompt protection; P can induce delayed immune memory

D) P and Q both supply antibodies without inducing memory

Q11

+4 / -1

A diploid locus is PCR-amplified to a 900 bp product. Allele R has one restriction site, giving 600 bp and 300 bp fragments after complete digestion; allele r has no site. The sample lane is shown. Assume complete digestion, equal PCR amplification per allele, equal DNA recovery, no other fragments and fluorescence proportional to DNA mass. What fraction of the sample-lane fluorescence should occur in the 900 bp band?

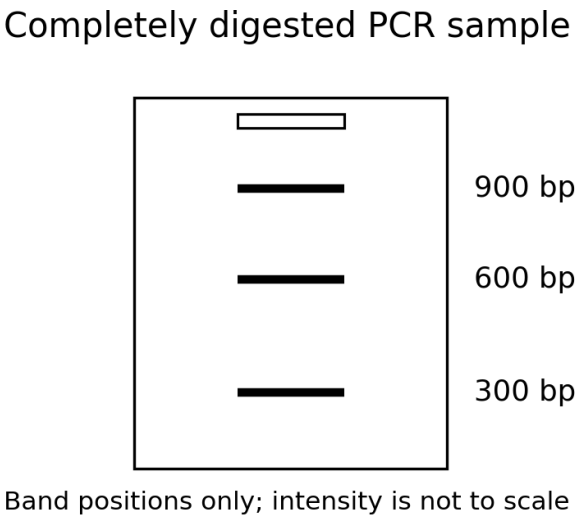


Figure for Q11

A) One-third

B) One-sixth

C) One-half

D) Two-thirds

Q12

+4 / -1

Viable muscle-cell preparations A and B have equal basal glucose uptake. A reagent S activates an intracellular step downstream of insulin reception, without binding insulin receptors. Uptake measurements are:

Treatment	A (units)	B (units)
None	2	2
Insulin	8	2
S	8	8

Insulin binding to the cell surface is normal in both preparations. Which conclusion is best supported?

A) B cannot take up glucose through any membrane pathway

B) B fails because its cells cannot synthesise insulin

C) B's insulin receptors cannot bind the supplied hormone

D) B has a defect after binding but before the step activated by S

Q13

+4 / -1

A fully penetrant autosomal-recessive disorder is caused by allele a. An unaffected person has an affected sibling and two unaffected parents; assume the parents are Aa × Aa. This person has a child with an unrelated, unaffected partner from a Hardy–Weinberg population where the frequency of a is 0.20. Assume random mating, no new mutation and independent family origins. What is the probability that the child is affected?

A) 1/12

B) 1/18

C) 1/24

D) 1/6

Q14

+4 / -1

Two resting frogs are fully submerged at the same temperature without lung ventilation. Their intact, moist skin has equal area and thickness, and skin blood flow is maintained.

Frog	Dissolved oxygen in water (relative units)	Oxygen uptake (relative units)
P	8	4
Q	4	2

Assume dissolved oxygen immediately inside the respiratory surface is negligible in both. Which interpretation and prediction best fit these observations?

A) Skin diffusion fits the data; lowering P's water oxygen to 4 should lower its uptake

B) Lung ventilation fits the data; raising Q's skin thickness should raise its uptake

C) Equal skin area requires equal uptake, regardless of the oxygen gradient

D) Uptake is independent of water oxygen if skin blood flow stays constant

Q15

+4 / -1

An initially neutral allele has frequency 0.50 in each of eight isolated experimental populations. Four populations are very small and four are very large. After many generations, the small populations have allele frequencies 0.00, 0.10, 0.90 and 1.00; the large populations remain near 0.50. Mutation, migration and fitness differences are excluded. Which explanation best fits the replicate outcomes?

A) Directional selection favours the allele equally in every population

B) The allele must have become beneficial in the two populations where it increased

C) Random sampling of alleles has stronger effects in the small populations

D) Hardy–Weinberg proportions require allele frequencies to change identically in all populations

Chemistry

Questions 16-30 · 15 questions · 60 marks

Q16

+4 / -1

An isolated carbonate ion, $\mathrm{CO_3}^{2-}$, is described using its three equivalent principal Lewis resonance structures. Which statement about its three carbon–oxygen bonds is correct?

A) Unequal bonds; one permanent double bond

B) Unequal bonds; switching double-bond location

C) Equivalent bonds; average bond order 2

D) Equivalent bonds; average bond order 4/3

Q17

+4 / -1

Compare octahedral $[\mathrm{Co(NH_3)_6}]^{3+}$ and $[\mathrm{Co(CN)_6}]^{3-}$. Assume the observed absorption in each comparison is the transition associated with its octahedral splitting Δ_o. Cyanide produces the larger splitting. Which statement follows?

A) The cyanide complex absorbs at shorter wavelength, while cobalt is +3 in both complexes

B) The cyanide complex absorbs at longer wavelength, while cobalt is +3 in both complexes

C) The two complexes must absorb at the same wavelength because both have six ligands

D) The cobalt oxidation states are +3 and −3, respectively, so ligand strength is irrelevant

Q18

+4 / -1

For a reaction, $\Delta H = +24.0\ \mathrm{kJ\,mol^{-1}}$ and $\Delta S = +80.0\ \mathrm{J\,K^{-1}mol^{-1}}$, independent of temperature over the range considered. Which pair gives ΔG at 200 K and 400 K, respectively?

A) −8.0 and +8.0 kJ mol−1

B) +40.0 and +56.0 kJ mol−1

C) +8.0 and −8.0 kJ mol−1

D) +23.984 and +23.968 kJ mol−1

Q19

+4 / -1

A mixture contains only propanal and propanone. Separate complete tests on identical portions of the mixture give 2.16 g of silver with Tollens reagent and 3.94 g of iodoform in the iodoform test. Propanal gives no iodoform under these conditions. Take $M(\mathrm{Ag}) = 108\ \mathrm{g\,mol^{-1}}$ and $M(\mathrm{CHI_3}) = 394\ \mathrm{g\,mol^{-1}}$. What is the mole fraction of propanone in the mixture?

A) 1/3

B) 1/2

C) 2/3

D) 1/4

Q20

+4 / -1

A zinc–copper galvanic cell has $E^\circ_{\mathrm{cell}} = 1.10\ \mathrm{V}$ and $[\mathrm{Zn^{2+}}] = 1.0\ \mathrm{M}$. In the copper half-cell, solid $\mathrm{Cu(OH)_2}$ is present and a buffer maintains $[\mathrm{OH^-}] = 10^{-4}\ \mathrm{M}$. Given $K_{sp}(\mathrm{Cu(OH)_2}) = 10^{-20}$, estimate the cell emf. Treat activities as concentrations, neglect junction potentials, and take $2.303RT/F = 0.060\ \mathrm{V}$.

A) 1.46 V

B) 1.10 V

C) 0.38 V

D) 0.74 V

Q21

+4 / -1

A mixture contains only methylamine and aniline. It is treated with excess nitrous acid at 0–5 °C, then warmed after cold-stage gas evolution stops. The graph shows the cumulative nitrogen collected. Assume quantitative reactions, negligible gas loss and complete thermal decomposition of the aromatic diazonium salt on warming. Which initial amounts of methylamine and aniline, respectively, are consistent with the graph?

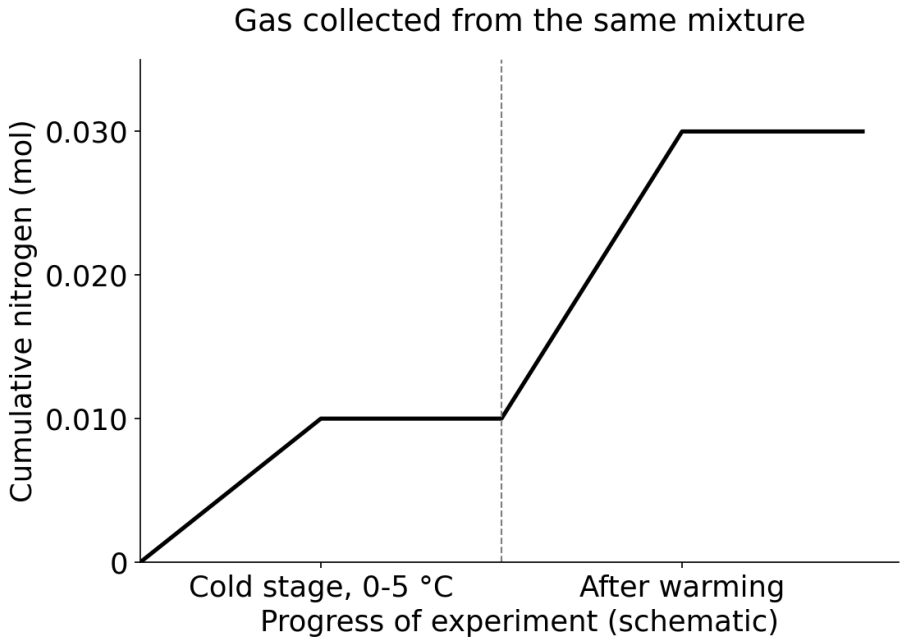


Figure for Q21

A) 0.030 mol; 0.010 mol

B) 0.020 mol; 0.010 mol

C) 0.010 mol; 0.020 mol

D) 0.010 mol; 0.030 mol

Q22

+4 / -1

Treat $\mathrm{Fe_3O_4}$ as containing only $\mathrm{Fe^{2+}}$, $\mathrm{Fe^{3+}}$ and $\mathrm{O^{2-}}$. What fraction of its iron ions must be in the +3 oxidation state?

A) 2/3

B) 1/3

C) 1/2

D) 1

Q23

+4 / -1

A first-order reaction is 25% complete after 10.0 min. How much additional time is required for the reaction to become 75% complete? Take $\ln(4/3) = 0.288$ and $\ln 3 = 1.099$.

A) 20.0 min

B) 38.2 min

C) 48.2 min

D) 30.0 min

Q24

+4 / -1

Racemic $\mathrm{CH_3COCH(CH_3)CH{=}CH_2}$ is treated with $\mathrm{NaBH_4}$, reducing only the ketone to an alcohol. Both faces of the carbonyl can react. After work-up, all the alcohol is acylated with ethanoyl chloride in pyridine. The terminal C=C bond survives, and no stereocentre subsequently changes configuration. What are the molecular formula and maximum number of distinct configurational stereoisomers of the final ester? Count all products accessible from both starting enantiomers; ignore conformers.

A) $\mathrm{C_8H_{12}O_2}$; 4

B) $\mathrm{C_8H_{14}O_2}$; 2

C) $\mathrm{C_8H_{14}O_2}$; 4

D) $\mathrm{C_8H_{14}O_2}$; 8

Q25

+4 / -1

The figure gives the connectivity of alkenes P, Q and R; it does not assign a particular geometrical configuration to Q. Which classification is correct? Ignore conformers and count geometrical isomerism only where each double-bond carbon has two different substituents.

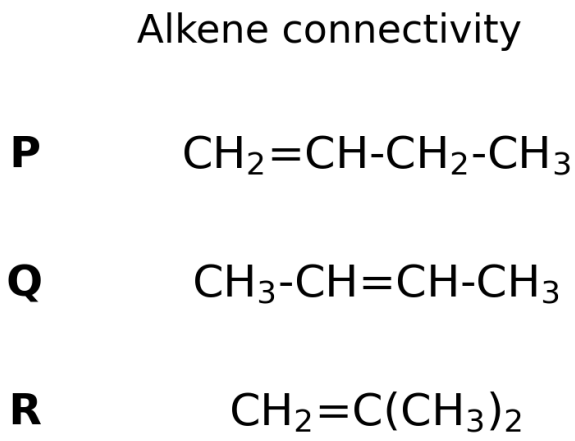


Figure for Q25

A) P and Q are chain isomers; P and R are position isomers; all three show E/Z isomerism

B) P and Q are position isomers; P and R are chain isomers; P and Q show E/Z isomerism

C) P and Q are chain isomers; P and R are position isomers; only R shows E/Z isomerism

D) P and Q are position isomers; P and R are chain isomers; only Q shows E/Z isomerism

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Q26

+4 / -1

Two solutions at temperature T initially have equal hydrostatic pressure and are separated by an ideal membrane permeable only to water. The left side contains 0.10 M sucrose. The right side contains 0.060 M $\mathrm{CaCl_2}$, of which 75% is dissociated; assume ideal behaviour otherwise. Which initial water-flow direction and extra pressure would prevent that flow? Use R in L atm mol−1 K−1.

A) Water flows right to left; apply an extra 0.040RT atm on the left

B) Water flows left to right; apply an extra 0.050RT atm on the right

C) Water flows left to right; apply an extra 0.080RT atm on the right

D) No initial flow occurs because both solutes have the same mass concentration

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Q27

+4 / -1

A 1.90 g mixture contains only anhydrous $\mathrm{Na_2CO_3}$ and $\mathrm{NaHCO_3}$. Heating completely decomposes the bicarbonate and leaves 1.59 g of anhydrous residue. What was the mass percentage of sodium bicarbonate in the original mixture? Use molar masses 106 and 84 g/mol for sodium carbonate and sodium bicarbonate, respectively; the only volatile products are carbon dioxide and water.

A) 44.2%

B) 22.1%

C) 55.8%

D) 16.3%

Q28

+4 / -1

Consider the ions $\mathrm{S^{2-}}$, $\mathrm{Cl^-}$, $\mathrm{K^+}$ and $\mathrm{Ca^{2+}}$. Their atomic numbers are 16, 17, 19 and 20, respectively. Which order of ionic radii is expected for these isoelectronic ions?

A) $\mathrm{Ca^{2+} > K^+ > Cl^- > S^{2-}}$

B) $\mathrm{K^+ > Ca^{2+} > S^{2-} > Cl^-}$

C) $\mathrm{Cl^- > S^{2-} > Ca^{2+} > K^+}$

D) $\mathrm{S^{2-} > Cl^- > K^+ > Ca^{2+}}$

Q29

+4 / -1

Separate carbohydrate samples X, Y and Z give the results below. Hydrolysates are neutralised before testing. Positive results mean a reducing sugar is present; hydrolysis is complete.

Sample	Reducing-sugar test before hydrolysis	Products after hydrolysis
X	Negative	Glucose and fructose
Y	Positive	Glucose only
Z	Positive	Glucose and galactose

Which identification fits all observations?

A) X maltose; Y sucrose; Z lactose

B) X sucrose; Y lactose; Z maltose

C) X sucrose; Y maltose; Z lactose

D) X lactose; Y maltose; Z sucrose

Q30

+4 / -1

100 mL of 0.40 M ammonia is mixed with 100 mL of 0.10 M HCl. Then 0.020 mol more HCl is added in a small volume. Assuming quantitative neutralisation followed by weak-base equilibrium, what is the final pH? Take $K_b(\mathrm{NH_3}) = 10^{-5}$, $pK_w = 14.00$ and $\log 3 = 0.477$. Activity effects may be neglected.

A) 8.52

B) 9.48

C) 5.00

D) 7.00

QUESTION PAPER / M

Mathematics

Questions 31-45 · 15 questions · 60 marks

Q31

+4 / -1

For $A = \begin{pmatrix} 3 & 1 \\ 1 & 1 \end{pmatrix}$, what is the sum of the diagonal entries of A^{-1}?

A) 4

B) 1/2

C) 2

D) -2

Q32

+4 / -1

Six distinct folders A, B, C, D, E and F are arranged in a row. A must be to the left of B, with exactly one folder between A and B. How many arrangements satisfy these conditions?

A) 48

B) 96

C) 192

D) 120

Q33

+4 / -1

A function $f : \mathbb{R} \to \mathbb{R}$ satisfies $f(x) + 2f(1 - x) = x^2$ for every real x. What is the minimum value of f on $[-1, 2]$?

A) 0

B) 2/3

C) $-1/3$

D) $-2/3$

Q34

+4 / -1

Two complex numbers z_1 and z_2 are exactly the solutions of $|z - 1| = |z + 1|$ and $|z - 2i| = \sqrt{5}$. What is $\mathrm{Im}(z_1)\,\mathrm{Im}(z_2)$?

A) -1

B) 1

C) -5

D) 4

Q35

+4 / -1

Evaluate $I = \int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x}\, dx$.

A) π/2

B) π²/2

C) π²/4

D) π²/8

Q36

+4 / -1

Point P has coordinates (1,2,3). Its reflection in the plane $x + y + z = 3$ is P′. What is the distance of P′ from the origin?

A) √14

B) √5

C) 2√3

D) √2

Q37

+4 / -1

Real numbers a and b are chosen so that $f(x) = |x^2 - 1| + a|x - 1| + b|x + 1|$ is differentiable at every real x. What is a+b?

A) 4

B) −4

C) −2

D) 0

Q38

+4 / -1

Two groups each contain four observations. Group A has mean 2 and population variance 1; group B has mean 6 and population variance 9. Here each variance uses the number of observations, not one less, as its denominator. What is the population variance of all eight observations together?

A) 9

B) 5

C) 13

D) 18

Q39

+4 / -1

Find $\lim_{x \to 0} \dfrac{\sqrt{1+3x} - \sqrt{1-x}}{x}$.

A) 4

B) 1

C) 0

D) 2

Q40

+4 / -1

Six distinct keys are assigned randomly, one each, to six numbered locks. Exactly one key is correct for each lock, and every one of the $6!$ assignments is equally likely. What is the probability that exactly two locks receive their correct keys?

A) 1/8

B) 1/48

C) 3/16

D) 1/2

Q41

+4 / -1

Using principal values of inverse tangent, evaluate $\tan^{-1} 2 + \tan^{-1} 3$.

A) $3\pi/4$

B) $-\pi/4$

C) $\pi/4$

D) $\tan^{-1} 5$

Q42

+4 / -1

For $x > 0$, a function satisfies $\dfrac{dy}{dx} = \dfrac{y}{x(1+x)}$ and y(1)=2. What is y(3)?

A) 3/2

B) 3

C) 6

D) 4

Q43

+4 / -1

For which real values of a does $x^2 - 2ax + a + 2 = 0$ have two real roots r1 and r2 satisfying $r_1 < 1 < r_2$?

A) a ≥ 3

B) a < −1 or a > 2

C) a > 3

D) a < 3

Q44

+4 / -1

The line of slope 1 through the focus of the parabola $y^2 = 4x$ meets it at P and Q. The tangents to the parabola at P and Q meet at T. What is the area of triangle PQT, in square units?

A) $4\sqrt{2}$

B) 8

C) $16\sqrt{2}$

D) $8\sqrt{2}$

Q45

+4 / -1

Unit vectors $\mathbf{a}$ and $\mathbf{b}$ make an angle of 60°. A parallelogram has adjacent side vectors $\mathbf{a} + 2\mathbf{b}$ and $\mathbf{a} - \mathbf{b}$. What is its area?

A) √3/2

B) 3√3/2

C) 3

D) 3√3/4

Physics

Questions 46-60 · 15 questions · 60 marks

Q46

+4 / -1

A particle moves along a straight line with the velocity-time graph shown. What are its average speed and average velocity, respectively, over the full 6 s interval?

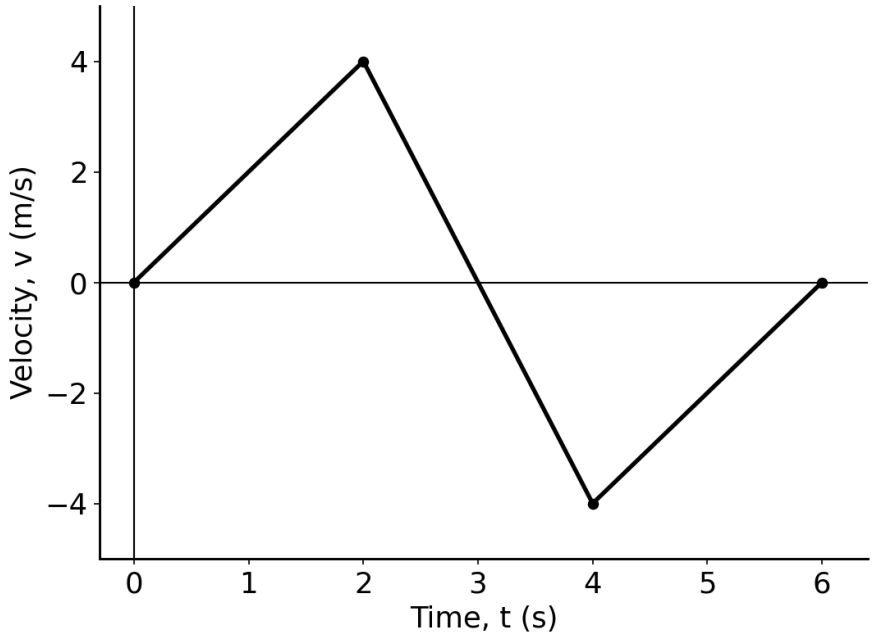


Figure for Q46

A) $2\,\mathrm{m\,s^{-1}},\ 0$

B) $0,\ 2\,\mathrm{m\,s^{-1}}$

C) $4/3\,\mathrm{m\,s^{-1}},\ 0$

D) $2\,\mathrm{m\,s^{-1}},\ 2\,\mathrm{m\,s^{-1}}$

Q47

+4 / -1

An empirical pressure model has the form $P = a\rho v^2 + bh$, where density is ρ, speed is v, height is h, and a is dimensionless. What are the dimensions of b?

A) $ML^{-1}T^{-2}$

B) LT^{-2}

C) $ML^{-2}T^{-2}$

D) ML^{-3}

Q48

+4 / -1

An incompressible, nonviscous fluid of density ρ flows steadily through a horizontal pipe. A wide section has radius r and speed v; a narrow section has radius r/2. What is the pressure in the wide section minus that in the narrow section?

A) $\frac{3}{2}\rho v^2$

B) $\frac{15}{2}\rho v^2$

C) $8\rho v^2$

D) $4\rho v^2$

Q49

+4 / -1

A horizontal disc has constant moment of inertia $2mR^2$ about its frictionless vertical axle. A point bead of mass m initially at radius R co-rotates with it at angular speed ω. A powered actuator attached to the disc moves the bead slowly along a radial groove to R/2. Initially and finally the bead has no radial velocity and co-rotates with the disc. The actuator has negligible inertia; there is no external torque about the axle and no mechanical dissipation. What are the final angular speed and the work supplied by the actuator to the disc-bead mechanical motion?

A) $3\omega/4,\ -mR^2\omega^2/2$

B) $4\omega/3,\ 0$

C) $\omega,\ mR^2\omega^2/2$

D) $4\omega/3,\ mR^2\omega^2/2$

Q50

+4 / -1

A cyclic heat engine absorbs 800 J from a reservoir at 500 K, delivers 300 J of work and rejects the remaining heat to a reservoir at 300 K. Which statement is correct?

A) These figures are consistent with an irreversible engine.

B) These figures describe a reversible engine.

C) These figures necessarily violate the first law of thermodynamics.

D) Its efficiency exceeds the Carnot limit for these reservoirs.

Q51

+4 / -1

A mass m stretches an ideal massless vertical spring by 10 cm at equilibrium. Let $\omega = \sqrt{k/m}$. The mass is initially 3 cm below equilibrium and is given a downward speed $v_0 = (0.04\ \text{m})\omega$. With no damping, what is the maximum tensile force in the spring?

A) mg/2

B) 3mg/2

C) 17mg/10

D) mg

Q52

+4 / -1

An unknown tuning fork produces 4 beats per second with the fundamental of a stretched string at 512 Hz and tension T. The string's vibrating length and linear mass density remain constant. Reducing its tension to $(255/256)^2 T$ makes the beat frequency 2 Hz. What is the fork frequency, and what tension would make the beats vanish?

A) $506\ \text{Hz};\ (253/256)^2 T$

B) $510\ \text{Hz};\ (255/256)^2 T$

C) $508\ \text{Hz};\ (127/128)^2 T$

D) $516\ \text{Hz};\ (129/128)^2 T$

Q53

+4 / -1

A positive charge q moves along the x axis under only the electric force. The field component is shown; $E_0 > 0$ and $a > 0$. At x = 0 the charge moves towards increasing x with kinetic energy $5qE_0a/4$. At what coordinate does it first come to rest?

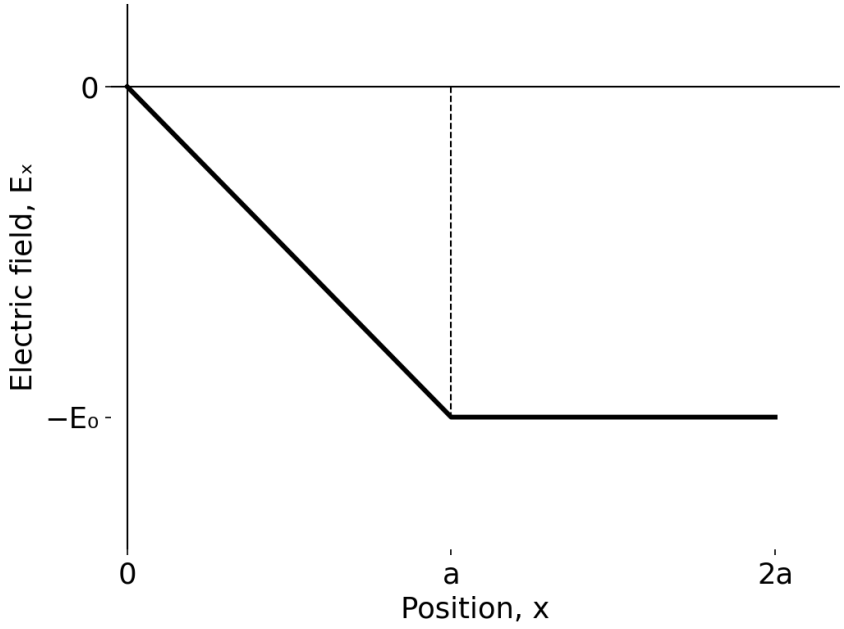


Figure for Q53

A) 5a/4

B) 3a/2

C) 2a

D) 7a/4

Q54

+4 / -1

In the circuit shown, the top rail is 12 V above the bottom rail. The source and ammeter are ideal; all other wires have negligible resistance. What current flows through the ammeter?

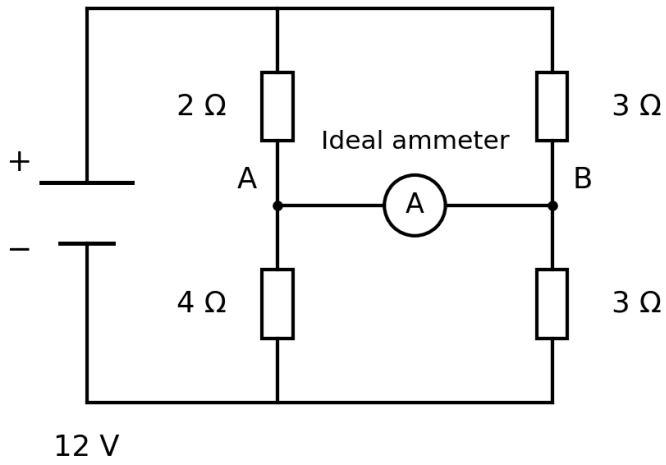


Figure for Q54

A) 2 A, from A to B

B) $12/17$ A, from A to B

C) $12/17$ A, from B to A

D) Zero

Q55

+4 / -1

A nucleus initially has mass number 238 and atomic number 92. It undergoes one alpha decay followed by two beta-minus decays, with no other nuclear transformation. What are its final mass number and atomic number, respectively?

A) 234, 92

B) 234, 90

C) 238, 94

D) 236, 92

Q56

+4 / -1

A conducting bar of mass m and length l spans two parallel conducting rails perpendicular to their length and slides down them. The rails are inclined at angle θ to the horizontal. A uniform magnetic field of magnitude B is perpendicular to the rail plane. The closed circuit has constant total resistance R. Neglect friction, inductance and all other drag. What is the terminal speed of the bar?

A) $mgR\cos\theta/(B^2l^2)$

B) $mgR\sin\theta/(2B^2l^2)$

C) $2mgR\sin\theta/(B^2l^2)$

D) $mgR\sin\theta/(B^2l^2)$

Q57

+4 / -1

A coin lies at the bottom of the two transparent, immiscible liquid layers shown. It is viewed from air close to the normal. Neglect container thickness. What is its apparent depth below the top liquid surface, and would reversing the order of the layers (keeping each layer's depth) change this apparent depth?

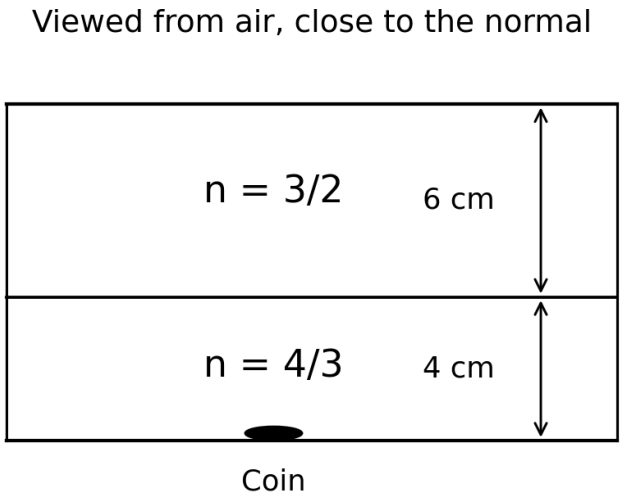


Figure for Q57

A) 10 cm; unchanged

B) 7 cm; changes

C) 7 cm; unchanged

D) $43/3$ cm; unchanged

Q58

+4 / -1

A series circuit contains an ideal resistor, inductor and capacitor driven by a sinusoidal source. The rms current is 2 A, and the rms voltage magnitudes across R, L and C are 8 V, 10 V and 4 V, respectively. What are the source rms voltage and average power supplied?

A) 10 V, 16 W

B) 22 V, 44 W

C) 10 V, 20 W

D) 14 V, 28 W

Q59

+4 / -1

An ideal diode and a resistor R form a half-wave rectifier driven by $v(t) = V_0 \sin \omega t$. There is no smoothing capacitor. What is the ratio of average power dissipated in R to the average power that the same sinusoidal source would dissipate in R if R were connected directly to it? Neglect source resistance and diode voltage drop.

A) $1/\sqrt{2}$

B) 1/2

C) 1/4

D) 1

Q60

+4 / -1

An ideal interferometer uses light of vacuum wavelengths 600 nm and 1000 nm. For each colour the two interfering beams are coherent and have equal amplitudes. At a reference point the geometric optical paths are initially equal, but one beam has an additional phase of π for each colour, so both colours are dark there. A plane glass slab is inserted normally in one path, replacing an equal thickness of air without otherwise changing the geometry. Its refractive index is 1.50 for both colours. What is the smallest positive slab thickness that makes the reference point an interference maximum for both colours?

A) 0.6 µm

B) 1.2 µm

C) 1.8 µm

D) 3.0 µm

Answer key

One correct option per question. Question and option order are fixed.

Biology

Q	Answer
1	**B**
2	**D**
3	**A**
4	**C**
5	**B**
6	**A**
7	**D**
8	**C**
9	**A**
10	**B**
11	**C**
12	**D**
13	**B**
14	**A**
15	**C**

Chemistry

Q	Answer
16	**D**
17	**A**
18	**C**
19	**B**
20	**D**
21	**C**
22	**A**
23	**B**
24	**C**
25	**D**
26	**B**
27	**A**
28	**D**
29	**C**
30	**A**

Mathematics

Q	Answer
31	**C**
32	**B**
33	**D**
34	**A**
35	**C**
36	**D**
37	**B**
38	**A**
39	**D**
40	**C**
41	**A**
42	**B**
43	**C**
44	**D**
45	**B**

Physics

Q	Answer
46	**A**
47	**C**
48	**B**
49	**D**
50	**A**
51	**B**
52	**C**
53	**D**
54	**B**
55	**A**
56	**D**
57	**C**
58	**A**
59	**B**
60	**D**

Reviewer checklist

Accuracy and single correct answer · Syllabus fit · Distractor quality
Diagram labels · Solution clarity · Difficulty and timed attempt

Reviewer: _____ Date: _____

Decision / notes: _____

Editorial difficulty mix: 9 easy / 42 medium / 9 hard. These labels require human review and are not official item statistics.

Biology

Method, reasoning and common errors

Q1

Biology · easy

Diversity In the Living World / Plant kingdom: bryophytes and pteridophytes; dominant generations

Answer: (B) X is a moss and Y is a fern.

Why: A moss has a dominant gametophyte and a dependent, attached sporophyte. In a fern the conspicuous, independent leafy plant is the sporophyte. Gamete versus spore production identifies the generations before the plants are assigned.

Why not the others:

(A) Reverses the characteristic dominance and dependence.

(C) Does not explain Y's dominant independent sporophyte.

(D) Does not explain X's dependent sporophyte.

Key point: Identify the generation by what it produces, then use its independence to distinguish the plant groups.

Q2

Biology · medium

Structural Organisation In Plants And Animals / Anatomy of flowering plants: monocot and dicot stems; open and closed bundles

Answer: (D) Q can establish a vascular cambium ring; P normally cannot.

Why: Q has open bundles, and its intrafascicular cambium can join interfascicular cambium during typical dicot-stem secondary growth. P describes the usual monocot stem with scattered closed bundles. The prediction concerns ordinary secondary growth, not exceptional monocot growth mechanisms.

Why not the others:

(A) Reverses the cambium evidence.

(B) Typical roots have radial vascular arrangement, not the described conjoint stem bundles.

(C) Q explicitly contains cambium and is therefore open.

Key point: Bundle position and cambium presence are separate observations; use both.

Q3

Biology · medium

Cell: Structure And Functions / Biomolecules: enzyme inhibition and effects of temperature

Answer: (A) Competitive inhibition is consistent with I, while H has lost functional enzyme.

Why: At low substrate, I slows the reaction, but excess substrate restores approximately the untreated maximum rate. That is the competitive-inhibition pattern. H remains near one-tenth of U at both conditions, consistent with persistent heat denaturation reducing active enzyme. These results support an interpretation rather than uniquely proving a molecular inhibitor mechanism.

Why not the others:

(B) Irreversible loss would not be rescued by excess substrate; H is not restored.

(C) I never exceeds U's maximum, and H's low maximum does not show improved affinity.

(D) Substrate excess does not rescue H.

Key point: Distinguish substrate rescue from recovery of the enzyme's maximum activity.

Q4

Biology · medium

Genetics And Evolution / Principles of inheritance and variation: linkage, recombination and gamete probabilities

Answer: (C) 0.64%.

Given: 1000 test-cross offspring; ab occurs 80 times.

Formula/Principle: A test-cross reads out the heterozygote's gametes. The aabb offspring must receive ab from each parent.

Step 1: The abundant classes Ab and aB are parental; the recombination fraction is (80 + 80)/1000 = 0.16.

Step 2: Each parent makes ab gametes with probability 80/1000 = 0.08.

Step 3: Independent gamete union gives P(aabb) = 0.08 × 0.08 = 0.0064 = 0.64%.

Common mistake: (A) is ten times too large; (B) squares a parental-gamete frequency; (D) treats the linked two-locus outcome as a single-gene recessive event.

Q5

Biology · medium

Human Physiology / Excretory products and their elimination: ADH, urine concentration and solute excretion

Answer: (B) Solute output is unchanged and water permeability increases.

Why: Daily osmole excretion equals urine volume times osmolarity: before, 2.0 × 300 = 600 mOsm/day; after, 0.75 × 800 = 600 mOsm/day. The same solute leaves in less water. With filtration and the medullary gradient held constant, increased collecting-duct water permeability, consistent with increased ADH action, explains the observations.

Why not the others:

(A) Lower volume does not imply lower solute output, and decreased permeability predicts the opposite water response.

(C) Confuses concentration with daily solute output.

(D) Gets the output right but reverses the permeability response.

Key point: Compare excretion rates, not concentration alone; these data are consistent with ADH action but do not prove a unique hormone concentration.

Q6

Biology · medium

Ecology / Ecosystem: gross/net productivity and trophic energy budgets

Answer: (A) 4%.

Given: Plant gross production 2400; plant respiration 900; herbivore assimilation 180 and respiration 120, all in kJ m−2 day−1.

Formula/Principle: NPP = GPP − producer respiration; net secondary production = assimilation − consumer respiration.

Step 1: Plant NPP = 2400 − 900 = 1500 kJ m−2 day−1.

Step 2: New herbivore biomass stores 180 − 120 = 60 kJ m−2 day−1.

Step 3: The requested percentage is (60/1500) × 100 = 4%.

Common mistake: (B) uses assimilation/GPP; (C) uses assimilation/NPP before consumer respiration; (D) uses consumer production/GPP instead of NPP.

Q7

Biology · easy

Cell: Structure And Functions / Cell cycle and cell division: chromosome number and DNA replication

Answer: (D) 12 chromosomes and 4C DNA.

Why: S phase duplicates each DNA molecule to create sister chromatids. It doubles DNA content but does not yet separate centromeres into daughter chromosomes. The cell remains diploid in G2.

Why not the others:

(A) Counts chromatids as separate chromosomes before centromere separation.

(B) Omits DNA replication.

(C) Increases chromosome count while failing to double DNA.

Key point: DNA quantity and chromosome number need not change together.

Q8

Biology · medium

Reproduction / Sexual reproduction in flowering plants: chromosome contributions in double fertilisation

Answer: (C) Zygote 36 with 1:2 contributions; endosperm nucleus 48 with 1:1 contributions.

Why: The female produces an egg and each polar nucleus with 12 chromosomes. Male meiosis produces gametes with 24. Syngamy gives 12 + 24 = 36, with maternal:paternal contribution 12:24. Triple fusion gives 12 + 12 + 24 = 48, with contribution 24:24. The usual 2:1 endosperm ratio assumes parents of the same ploidy and cannot simply be transferred here.

Why not the others:

(A) Assumes both parents are diploid.

(B) Applies the usual 2:1 ratio despite the specified gamete counts.

(D) Confuses the two fusion events.

Key point: Count the chromosomes in each participating nucleus before using a memorised ploidy ratio.

Q9

Biology · medium

Plant Physiology / Photosynthesis in higher plants: chemiosmosis and ATP synthesis

Answer: (A) Transient ATP synthesis is possible without light-dependent NADPH formation.

Why: The lumen has higher proton concentration than the outside. Proton movement down this gradient through ATP synthase can power phosphorylation outside the vesicle. The finite imposed gradient dissipates, so ATP production is transient. A gradient by itself does not supply the light-driven electron transfer that reduces NADP+.

Why not the others:

(B) Exchanges the roles of electron transfer and the proton gradient.

(C) Ignores the energy already stored in the imposed gradient.

(D) Reverses the downhill proton direction and invents light-driven reduction in darkness.

Key point: Light normally establishes the gradient, but an imposed gradient can temporarily power ATP synthesis.

Q10

Biology · easy

Biology In Human Welfare / Human health and disease: passive and active immunity

Answer: (B) P acts sooner; Q can induce memory after a delay.

Why: Ready-made antitoxin antibodies supply passive protection while they persist. A toxoid acts as antigen and stimulates the recipient's own adaptive response, including memory formation. This is a comparison of mechanisms, not a treatment recommendation.

Why not the others:

(A) Neither preparation supplies ready-made memory cells.

(C) Interchanges passive protection and active memory formation.

(D) A toxoid supplies antigen, not ready-made antibodies.

Key point: Receiving antibodies and generating one's own immune response are different processes.

Q11

Biology · hard

Biotechnology / Biotechnology: principles and processes; restriction digestion, PCR and electrophoresis

Answer: (C) One-half.

Given: The sample contains bands at 900, 600 and 300 bp; complete digestion and equal amplification are stipulated.

Formula/Principle: R contributes two cut fragments; r contributes one uncut fragment. Fluorescence is proportional to base-pair mass, not simply to the number of bands.

Step 1: All three bands imply an Rr sample under the stated assumptions.

Step 2: For one equally amplified copy of each allele, band masses are proportional to 900, 600 and 300.

Step 3: The 900 bp fraction is 900/(900 + 600 + 300) = 1/2.

Common mistake: (A) gives every band equal mass; (B) is the 300 bp mass fraction; (D) does not conserve the equal 900 bp contribution from the two alleles.

Q12

Biology · medium

Human Physiology / Chemical coordination and integration: insulin action and feedback

Answer: (D) A defect between binding and the S-activated step is supported.

Why: Normal surface binding excludes failure of insulin binding as the explanation. Rescue by S shows that uptake machinery downstream of the bypass can respond. The failure of insulin alone therefore localises a functional defect to the interval between reception and that bypassed step; it does not identify a particular protein.

Why not the others:

(A) Both basal uptake and the S response contradict total loss of uptake.

(B) Externally supplied insulin tests response rather than cellular insulin synthesis.

(C) Contradicts the binding assay.

Key point: Combine the binding measurement and bypass control; neither measurement alone localises the defect.

Q13

Biology · hard

Genetics And Evolution / Principles of inheritance and variation; Hardy–Weinberg principle and conditional carrier risk

Answer: (B) 1/18.

Given: The first parent is an unaffected child of Aa × Aa; the partner is unaffected and q = 0.20.

Formula/Principle: Condition genotype probabilities on being unaffected before multiplying transmission risks.

Step 1: The first person's carrier probability is (1/2)/(3/4) = 2/3.

Step 2: In the population, p = 0.80, so carrier frequency 2pq = 0.32 and unaffected frequency $1 - q^2 = 0.96$. The partner's conditional carrier probability is 0.32/0.96 = 1/3.

Step 3: Two carriers have an affected child with probability 1/4. Overall risk = (2/3)(1/3)(1/4) = 1/18.

Common mistake: (A) drops the first conditioning factor; (C) uses a 1/2 family carrier risk; (D) omits the partner's carrier uncertainty.

Q14

Biology · medium

Structural Organisation In Plants And Animals / Structural organisation in animals: amphibian-frog; respiration

Answer: (A) Cutaneous diffusion fits the gradient-dependent uptake.

Why: Under the stipulated equal surface area, thickness and maintained blood flow, the oxygen concentration difference is the changing diffusion condition. P has twice Q's external oxygen and twice its measured uptake. Lowering P's external oxygen reduces that driving difference and should reduce uptake. This comparison supports skin exchange; it does not establish that it meets every possible metabolic demand.

Why not the others:

(B) Lung ventilation is excluded; increasing thickness would impede diffusion.

(C) Ignores the concentration gradient.

(D) Holding blood flow constant does not eliminate dependence on the external gradient.

Key point: Respiratory exchange depends on both surface properties and a maintained concentration difference.

Q15

Biology · medium

Genetics And Evolution / Evolution: genetic drift, selection and population size

Answer: (C) Genetic drift is stronger in the small populations.

Why: With selection, migration and mutation excluded, finite sampling of parental alleles can change allele frequency. Opposite directions among replicate small populations, including loss and fixation, fit drift. Large populations show smaller sampling fluctuations.

Why not the others:

(A) Predicts a common selected direction and contradicts the controls.

(B) Infers a fitness advantage that the design excludes.

(D) Confuses genotype proportions under an ideal model with identical realised allele-frequency trajectories.

Key point: Replicate variability and population size can distinguish drift from a consistent selective advantage.

Chemistry

Method, reasoning and common errors

Q16

Chemistry · easy

Chemical Bonding and Molecular Structure / Kössel-Lewis structures, resonance and bond parameters

Answer: (D) Equivalent bonds with average bond order $4/3$.

Why: Each principal contributor contains one C=O bond and two C–O bonds. The resonance hybrid delocalises the bonding across equivalent oxygen positions; it is not a molecule switching between drawings. The model's average is $(2+1+1)/3 = 4/3$.

Why not the others:

(A) Incorrectly localises the double bond.

(B) Treats resonance contributors as separately interconverting molecules.

(C) Gives the model six bond-order units instead of four.

Key point: Resonance drawings are alternative representations of one delocalised structure.

Q17

Chemistry · medium

Coordination compounds / Bonding in coordination compounds: ligand-field splitting and absorption

Answer: (A) Shorter wavelength for the cyanide complex; Co is +3 in both.

Why: In the stated transition model, $\Delta_o = h\nu = hc/\lambda$. A larger splitting therefore requires higher frequency and shorter wavelength. Ammonia is neutral. In the cyanide complex, $x + 6(-1) = -3$, giving $x = +3$.

Why not the others:

(B) Reverses the energy–wavelength relationship.

(C) Coordination number alone does not fix splitting.

(D) Mistakes the complex charge for the oxidation state of the central metal.

Key point: Determine metal oxidation state by charge balance, and spectral energy by ligand-field splitting.

Q18

Chemistry · medium

Thermodynamics / Spontaneity and Gibbs energy change

Answer: (C) $+8.0$ and $-8.0\ \mathrm{kJ\,mol^{-1}}$.

Given: $\Delta H = 24.0\ \mathrm{kJ\,mol^{-1}}$; $\Delta S = 0.0800\ \mathrm{kJ\,K^{-1}mol^{-1}}$.

Formula/Principle: $\Delta G = \Delta H - T\Delta S$.

Step 1: At 200 K, $\Delta G = 24.0 - 200(0.0800) = +8.0\ \mathrm{kJ\,mol^{-1}}$.

Step 2: At 400 K, $\Delta G = 24.0 - 400(0.0800) = -8.0\ \mathrm{kJ\,mol^{-1}}$.

Step 3: The sign changes at $T = \Delta H/\Delta S = 300\ \mathrm{K}$.

Common mistake: (A) reverses the subtraction; (B) adds the entropy term; (D) mixes joules and kilojoules in the wrong direction.

Q19

Chemistry · medium

Aldehydes, ketones, and carboxylic acids / Chemical reactions: Tollens and iodoform tests; mixture analysis

Answer: (B) $1/2$.

Given: Identical portions are tested, so the inferred amounts can be compared directly.

Formula/Principle: One mole of aldehyde reduces two moles of silver ions; one mole of this methyl ketone gives one mole of iodoform.

Step 1: Silver amount is $2.16/108 = 0.0200\ \mathrm{mol}$, so propanal amount is $0.0200/2 = 0.0100\ \mathrm{mol}$.

Step 2: Propanone amount is $3.94/394 = 0.0100\ \mathrm{mol}$.

Step 3: $x_{\mathrm{propanone}} = 0.0100/(0.0100 + 0.0100) = 1/2$.

Common mistake: (A) treats one silver atom as one aldehyde molecule; (C) introduces an extra factor of two in the ketone amount; (D) divides the ketone fraction by two without a stoichiometric basis.

Q20

Chemistry · hard

Electrochemistry / Nernst equation coupled to solubility equilibrium

Answer: (D) $0.74\ \mathrm{V}$.

Given: The copper-ion concentration is controlled by precipitation equilibrium, not fixed at 1 M.

Formula/Principle: $K_{sp} = [\mathrm{Cu^{2+}}][\mathrm{OH^-}]^2$; for $\mathrm{Zn} + \mathrm{Cu^{2+}} \longrightarrow \mathrm{Zn^{2+}} + \mathrm{Cu}$, $n = 2$.

Step 1: $[\mathrm{Cu^{2+}}] = 10^{-20}/(10^{-4})^2 = 10^{-12}\ \mathrm{M}$.

Step 2: $Q = [\mathrm{Zn^{2+}}]/[\mathrm{Cu^{2+}}] = 10^{12}$.

Step 3: $E = 1.10 - (0.060/2)\log(10^{12}) = 1.10 - 0.36 = 0.74\ \mathrm{V}$.

Common mistake: (A) reverses the Nernst sign; (B) assumes standard concentrations; (C) omits the two-electron divisor.

Q21

Chemistry · medium

Organic compounds containing nitrogen / Amines and diazonium salts: interpreting staged nitrogen evolution

Answer: (C) 0.010 mol methylamine and 0.020 mol aniline.

Why: Primary aliphatic diazonium intermediates from methylamine decompose in the cold, giving one mole of nitrogen per mole of amine. Aniline forms a diazonium salt stable at 0–5 °C; its nitrogen evolves on warming. The first plateau therefore measures 0.010 mol methylamine. The additional gas is the difference between cumulative plateaus, 0.030 − 0.010 = 0.020 mol, and measures aniline.

Why not the others:

(A) Treats the final cumulative amount as methylamine.

(B) Reverses the two amines.

(D) Treats the final cumulative amount as additional warm-stage gas.

Key point: Use the chemical stability distinction and subtract cumulative readings before assigning the second component.

Q22

Chemistry · easy

Redox Reactions / Oxidation numbers and charge balance in mixed-valence compounds

Answer: (A) $2/3$.

Why: Four oxide ions contribute charge −8. If x of the three iron ions are Fe(III), the others are Fe(II), so $3x + 2(3 - x) = 8$. Thus $x = 2$, giving two Fe(III) ions out of three iron ions. The fractional average oxidation number does not imply that each individual iron ion has oxidation state 8/3.

Why not the others:

(B) Is the Fe(II) fraction.

(C) Does not supply total charge +8.

(D) Would supply +9.

Key point: An average oxidation number can represent a mixture of integer oxidation states.

Q23

Chemistry · medium

Chemical Kinetics / Integrated first-order rate equation and fractional conversion

Answer: (B) Approximately $38.2\ \text{min}$ additional time.

Given: The first interval leaves 75% of the starting reactant; the target leaves 25%.

Formula/Principle: For first order, $kt = \ln([R]_i/[R]_f)$.

Step 1: $k = \ln(1/0.75)/10 = 0.0288\ \text{min}^{-1}$.

Step 2: The additional interval takes concentration from 0.75 to 0.25 of the original, a ratio of 3.

Step 3: $\Delta t = \ln 3/k = 1.099/0.0288 \approx 38.2\ \text{min}$.

Common mistake: (A) assumes constant-rate percentage loss; (C) is the total time from the beginning; (D) confuses a remaining-fraction ratio with a number of equal time intervals.

Q24

Chemistry · hard

Aldehydes, ketones, and carboxylic acids / Carbonyl reduction and selective functional-group transformations; alcohol acylation

Answer: (C) $\text{C}_8\text{H}_{14}\text{O}_2$, with four possible configurational stereoisomers.

Why: The starting ketone has formula $\text{C}_6\text{H}_{10}\text{O}$ and one stereocentre at the carbon bearing the methyl branch. Ketone reduction adds two hydrogens to give $\text{C}_6\text{H}_{12}\text{O}$ and creates a second stereocentre at the former carbonyl carbon. Alcohol acylation replaces its hydroxyl hydrogen with COCH_3, a net addition of $\text{C}_2\text{H}_2\text{O}$, yielding $\text{C}_8\text{H}_{14}\text{O}_2$. There are two independent stereocentres and no internal symmetry, so all four configurations are distinct. The terminal alkene has two identical hydrogens on one carbon and supplies no E/Z pair.

Why not the others:

(A) Omits the two hydrogens gained in carbonyl reduction.

(B) Counts only one stereocentre, or only products from one starting enantiomer.

(D) Incorrectly doubles the count for E/Z isomerism of a terminal alkene.

Key point: Track elemental changes, newly created stereocentres and the eligibility for geometric isomerism separately.

Q25

Chemistry · medium

Organic Chemistry – Some Basic Principles and Techniques / Structural and geometrical isomerism: interpreting alkene connectivity

Answer: (D) P/Q are position isomers, P/R are chain isomers, and only Q has E/Z forms.

Why: P is but-1-ene and Q is but-2-ene: the same unbranched carbon skeleton has the double bond at different positions. R is 2-methylpropene and has a branched skeleton, so it is a chain isomer of P. In Q, each double-bond carbon bears H and methyl, allowing geometrical isomerism. P and R each have a terminal CH2 group with two identical substituents on one double-bond carbon, preventing E/Z isomerism. All three have molecular formula C4H8.

Why not the others:

(A) Reverses the structural categories and counts ineligible terminal alkenes.

(B) Incorrectly grants E/Z isomerism to P.

(C) Reverses both the structural categories and E/Z eligibility.

Key point: Read the carbon skeleton before locating the double bond, then check both ends of the double bond for E/Z eligibility.

Q26

Chemistry · medium

Solutions / Osmotic pressure, dissociation and the van't Hoff factor

Answer: (B) Left to right; extra pressure $0.050RT\ \text{atm}$ on the right.

Given: $\alpha = 0.75$ for a salt yielding three ions on full dissociation.

Formula/Principle: $i = 1 + (3-1)\alpha$ and $\Pi = iCRT$.

Step 1: $i = 1 + 2(0.75) = 2.50$. Effective particle concentration on the right is $2.50(0.060) = 0.150\ \text{mol}\,\text{L}^{-1}$.

Step 2: The left has 0.100 M particles, so water moves to the right.

Step 3: $\Delta\Pi = (0.150 - 0.100)RT = 0.050RT\ \text{atm}$, applied on the right.

Common mistake: (A) compares formula-unit concentrations only; (C) assumes full dissociation; (D) substitutes mass concentration for particle concentration.

Q27

Chemistry · medium

Some Basic Concepts of Chemistry / Mole concept and stoichiometry: decomposition followed by neutralisation

Answer: (A) 44.2%.

Given: Initial mass 1.90 g; final mass 1.59 g; lost mass 0.31 g.

Formula/Principle: $2\,\mathrm{NaHCO_3} \longrightarrow \mathrm{Na_2CO_3} + \mathrm{CO_2} + \mathrm{H_2O}$. For 168 g bicarbonate, the volatile mass is 44 + 18 = 62 g.

Step 1: Initial bicarbonate mass is $0.31(168/62) = 0.84$ g.

Step 2: Its original mass percentage is $100(0.84/1.90) = 44.2\%$, to three significant figures. The remaining 1.06 g carbonate plus 0.53 g newly formed carbonate gives the observed 1.59 g residue.

Common mistake: (B) loses the coefficient two in bicarbonate decomposition; (C) reports the original carbonate percentage; (D) treats all lost mass as bicarbonate.

Q28

Chemistry · medium

Classification of Elements and Periodicity in Properties / Electronic configurations and ionic-radius trends in isoelectronic species

Answer: (D) $\mathrm{S^{2-}} > \mathrm{Cl^{-}} > \mathrm{K^{+}} > \mathrm{Ca^{2+}}$ in ionic radius.

Why: Each ion has 18 electrons. With the electron count and occupied-shell pattern the same, the increasing nuclear charge pulls the electron cloud in more strongly. Nuclear charge rises from S to Ca, so radius decreases in that order.

Why not the others:

(A) Reverses the nuclear-charge trend.

(B) Does not follow the common-electron-count comparison.

(C) Reverses both the anion pair and the cation pair.

Key point: For an isoelectronic series, compare nuclear charge before applying neutral-atom size trends.

Q29

Chemistry · medium

Biomolecules / Carbohydrates: non-reducing disaccharides, hydrolysis and reducing sugars

Answer: (C) X sucrose, Y maltose, Z lactose.

Why: Sucrose is non-reducing and hydrolyses to glucose and fructose. Maltose is reducing and gives two glucose molecules. Lactose is reducing and gives glucose and galactose. X is therefore distinguished both by its initial test and its products; the identities of Y and Z follow from their different constituent monosaccharides.

Why not the others:

(A) Assigns a negative test to reducing maltose and glucose-only hydrolysis to sucrose.

(B) Interchanges the glucose-only and glucose/galactose disaccharides.

(D) Assigns the wrong reducing behaviour and products to X and Z.

Key point: Combine the intact molecule's reducing behaviour with the identities of its hydrolysis products.

Q30

Chemistry · hard

Equilibrium / Ionic equilibrium: neutralisation and a weak-base/conjugate-acid buffer

Answer: (A) Approximately 8.52.

Given: Initial ammonia amount $0.100(0.40) = 0.040\ \text{mol}$; first HCl amount $0.100(0.10) = 0.010\ \text{mol}$.

Formula/Principle: Perform strong-acid neutralisation first, then use $pOH = pK_b + \log([\text{NH}_4{}^+]/[\text{NH}_3])$.

Step 1: After the first addition, 0.030 mol ammonia remains and 0.010 mol ammonium is present.

Step 2: The next 0.020 mol HCl leaves 0.010 mol ammonia and produces a total 0.030 mol ammonium; no strong acid remains in excess.

Step 3: Both buffer species share the same final volume, so their concentration ratio is $0.030/0.010 = 3$.

Step 4: $pOH = 5.000 + 0.477 = 5.477$; $pH = 14.000 - 5.477 = 8.523$.

Common mistake: (B) uses the buffer ratio before the second addition; (C) mistakes pKb for pH; (D) assumes partial neutralisation automatically gives a neutral solution.

Mathematics

Method, reasoning and common errors

Q31

Mathematics · easy

Matrices & determinants / Adjoint and inverse of a matrix

Answer: (C) 2.

Given: $A = \begin{pmatrix} 3 & 1 \\ 1 & 1 \end{pmatrix}$.

Formula/Principle: $A^{-1} = \operatorname{adj}(A) / \det A$.

Step 1: $\det A = 3(1) - 1(1) = 2 \neq 0$.

Step 2: $A^{-1} = \frac{1}{2} \begin{pmatrix} 1 & -1 \\ -1 & 3 \end{pmatrix}$.

Step 3: The diagonal sum is $1/2 + 3/2 = 2$.

Common mistake: (A) uses the original trace; (B) reports only the inverse-determinant factor; (D) introduces an unwarranted negative sign.

Q32

Mathematics · medium

Basic counting techniques / Permutations and Combinations: constrained arrangements

Answer: (B) 96.

Given: Six positions, with B exactly two positions to the right of A.

Formula/Principle: Choose the allowed positions of the constrained pair, then permute all remaining distinct folders.

Step 1: A can occupy position 1, 2, 3 or 4; B then occupies 3, 4, 5 or 6, respectively. There are four ordered position pairs.

Step 2: The other four folders occupy the remaining positions in $4! = 24$ ways.

Step 3: The total is $4 \times 24 = 96$.

Common mistake: (A) loses half the allowed position pairs; (C) incorrectly allows B to the left of A as well; (D) treats A and B as a single adjacent block, which they are not.

Q33

Mathematics · medium

Sets and Logic / Relations & functions: functional equations combined with minimisation

Answer: (D) $-2/3$.

Given: $f(x)+2f(1-x)=x^2$ for all real x.

Formula/Principle: Use the equation at x and at $1-x$ to eliminate the second function value.

Step 1: Substitution gives $f(1-x)+2f(x)=(1-x)^2$.

Step 2: Twice this new equation minus the original gives $3f(x)=2(1-x)^2-x^2=x^2-4x+2$.

Step 3: Hence $f(x)=((x-2)^2-2)/3$.

Step 4: The point x = 2 belongs to the closed interval and makes the square zero, so the minimum is $-2/3$.

Common mistake: (A) assumes a nonnegative right-hand side forces nonnegative f; (B) loses the constant's sign; (C) loses a factor during elimination.

Q34

Mathematics · medium

Complex numbers, linear and quadratic equations / Complex Numbers: geometric loci in the Argand plane

Answer: (A) -1.

Given: Write $z=x+iy$.

Formula/Principle: Equal distances from +1 and −1 define the perpendicular bisector of those points.

Step 1: $(x-1)^2+y^2=(x+1)^2+y^2$ gives x = 0.

Step 2: The other condition becomes $(y-2)^2=5$, with roots $y=2+\sqrt{5}$ and $2-\sqrt{5}$.

Step 3: Their product is $(2+\sqrt{5})(2-\sqrt{5})=4-5=-1$.

Common mistake: (B) takes the magnitude of the signed product; (C) ignores the circle's vertical displacement; (D) multiplies only the shared centre coordinate.

Q35

Mathematics · medium

Integration / Definite-integral symmetry and substitution

Answer: (C) $\pi^2/4$.

Given: The weight $g(x)=\sin x/(1+\cos^2 x)$ satisfies $g(\pi-x)=g(x)$.

Formula/Principle: Reflection of the limits gives $2I=\pi\int_0^{\pi} g(x)\,dx$.

Step 1: Replacing x by $\pi-x$ yields $I=\int_0^{\pi}(\pi-x)g(x)\,dx$; add this to the original integral.

Step 2: With $u=\cos x$, $\int_0^{\pi} g(x)\,dx=\int_{-1}^{1} du/(1+u^2)=\pi/2$.

Step 3: Thus $I=(\pi/2)(\pi/2)=\pi^2/4$.

Common mistake: (A) evaluates only the weight integral; (B) omits the division by two from reflection; (D) divides by two twice.

Q36

Mathematics · medium

Three-dimensional geometry / Cartesian equation of a plane; normal projection and reflection

Answer: (D) $\sqrt{2}$.

Given: A normal to the plane is $\mathbf{n} = (1, 1, 1)$.

Formula/Principle: Reflection in $\mathbf{n} \cdot \mathbf{r} = d$ is $\mathbf{r}' = \mathbf{r} - 2(\mathbf{n} \cdot \mathbf{r} - d)\mathbf{n}/|\mathbf{n}|^2$.

Step 1: Here $\mathbf{n} \cdot P - d = 6 - 3 = 3$ and $|\mathbf{n}|^2 = 3$.

Step 2: Therefore $P' = (1, 2, 3) - 2(1, 1, 1) = (-1, 0, 1)$.

Step 3: $|OP'| = \sqrt{(-1)^2 + 0^2 + 1^2} = \sqrt{2}$.

Common mistake: (A) uses the original point; (B) uses the projection foot (0,1,2), not the reflection; (C) uses twice the perpendicular plane distance, not the origin distance.

Q37

Mathematics · medium

Differentiation / Differentiability of absolute-value functions and one-sided derivatives

Answer: (B) -4.

Given: Only x = 1 and x = −1 can be cusp points.

Formula/Principle: A term $c|x - x_0|$ has a derivative jump of $2c$; differentiability requires the total jump to vanish.

Step 1: Near x = 1, $|x^2 - 1| = |x - 1||x + 1|$ has leading cusp $2|x - 1|$. The other nonsmooth contribution there is $a|x - 1|$, so $a = -2$.

Step 2: Similarly, the cusp at x = −1 has coefficient $2 + b$, so $b = -2$. Terms centred at the other point are smooth locally.

Step 3: $a + b = -4$.

Common mistake: (A) reinforces rather than cancels the cusps; (C) cancels just one cusp; (D) treats cancellation of the coefficients' sum as enough at two distinct locations.

Q38

Mathematics · medium

Statistics, Probability & Linear Programming / Statistics: mean and population variance of combined data

Answer: (A) 9.

Given: Equal group sizes; means 2 and 6; variances 1 and 9.

Formula/Principle: Combined variance equals the average within-group variance plus the variance of group means about the combined mean, with weights proportional to group sizes.

Step 1: Combined mean is $\mu = (2 + 6)/2 = 4$.

Step 2: Within-group contribution is $(1 + 9)/2 = 5$.

Step 3: Between-group contribution is $((2 - 4)^2 + (6 - 4)^2)/2 = 4$. Total variance is $5 + 4 = 9$.

Common mistake: (B) omits between-group variation; (C) adds both squared mean offsets without averaging; (D) fails to normalise the final variance by the combined population size correctly.

Q39

Mathematics · easy

Limit & continuity / Limits of algebraic functions: rationalisation

Answer: (D) 2.

Given: Both square roots are defined in a neighbourhood of zero.

Formula/Principle: Multiply the numerator and denominator by the conjugate.

Step 1: The numerator product becomes $(1+3x)-(1-x)=4x$.

Step 2: For nonzero x near zero, the quotient equals $4/(\sqrt{1+3x}+\sqrt{1-x})$.

Step 3: Letting x tend to zero gives $4/(1+1)=2$.

Common mistake: (A) omits the conjugate denominator; (B) loses a factor of two; (C) substitutes into the indeterminate numerator without treating the simultaneous zero denominator.

Q40

Mathematics · hard

Statistics, Probability & Linear Programming / Probability with permutations and inclusion-exclusion

Answer: (C) $3/16$.

Given: There are $6!=720$ equally likely assignments.

Formula/Principle: Choose the two correct assignments, then require that none of the other four is correct.

Step 1: Choose the two correctly assigned keys in $\binom{6}{2}=15$ ways.

Step 2: By inclusion-exclusion, the number of permutations of the remaining four keys with no fixed assignment is $D_4=4!-\binom{4}{1}3!+\binom{4}{2}2!-\binom{4}{3}1!+\binom{4}{4}0!=24-24+12-4+1=9$.

Step 3: Probability $=15(9)/720=3/16$.

Common mistake: (A) counts only four-cycles, omitting the three ways to form two disjoint swaps; (B) counts only the choice of the correct pair, omitting arrangements of the other keys; (D) allows all 4! assignments of the remainder and hence counts assignments with extra correct keys, some repeatedly.

Q41

Mathematics · medium

Trigonometric functions / Inverse trigonometric functions: addition formula and principal values

Answer: (A) $3\pi/4$.

Given: Each inverse tangent is in $(0,\pi/2)$, and each exceeds $\pi/4$.

Formula/Principle: $\tan(\alpha+\beta)=(\tan\alpha+\tan\beta)/(1-\tan\alpha\tan\beta)$.

Step 1: The tangent of the sum is $(2+3)/(1-6)=-1$.

Step 2: The sum is greater than $\pi/2$ and less than π, so it lies in the second quadrant.

Step 3: The second-quadrant angle with tangent -1 is $3\pi/4$.

Common mistake: (B) ignores the sum's quadrant; (C) loses the sign of the tangent denominator; (D) adds tangent arguments without the denominator correction.

Q42

Mathematics · medium

Differential equations / Solution by separation of variables with an initial condition

Answer: (B) 3.

Given: $y' = y/[x(1+x)]$, with $y(1) = 2$.

Formula/Principle: Separate variables and use $1/[x(1+x)] = 1/x - 1/(1+x)$.

Step 1: $\int dy/y = \int (1/x - 1/(1+x))\,dx$, so $y = Cx/(1+x)$ on the positive solution branch.

Step 2: $2 = C/2$, hence C = 4.

Step 3: $y(3) = 4(3)/4 = 3$.

Common mistake: (A) loses a factor when imposing the initial condition; (C) omits the final denominator change; (D) mistakes the integration constant for the requested function value.

Q43

Mathematics · medium

Complex numbers, linear and quadratic equations / Quadratic Equations: location of real roots

Answer: (C) $a > 3$.

Why: The monic quadratic opens upwards. A test point lies strictly between its two real roots exactly when the polynomial is negative there. At x = 1, the value is $1 - 2a + a + 2 = 3 - a$, so the required inequality is $3 - a < 0$. This also guarantees two distinct real roots because a continuous upward-opening quadratic is below zero at 1 and above zero for sufficiently large positive and negative x.

Why not the others:

(A) Includes a = 3, when 1 is a root rather than strictly between the roots.

(B) Only enforces distinct-real-root discriminant conditions.

(D) Reverses the sign test.

Key point: Root location requires more than a nonnegative discriminant.

Q44

Mathematics · hard

Coordinate geometry / Conic sections: focal chord of a parabola, tangents and triangle area

Answer: (D) $8\sqrt{2}$ square units.

Given: The focus is (1,0), so the chord is y = x − 1.

Formula/Principle: Parametrise the parabola by $(t^2, 2t)$; its tangent is $ty = x + t^2$. Triangle area is half of base times perpendicular height.

Step 1: Substituting the chord gives $t^2 - 2t - 1 = 0$, so $t_1 + t_2 = 2$, $t_1 t_2 = -1$, and $t_1 - t_2 = 2\sqrt{2}$.

Step 2: Subtracting the two tangent equations gives $y_T = t_1 + t_2 = 2$, then $x_T = t_1 t_2 = -1$.

Step 3: The endpoints differ by $4\sqrt{2}$ in both coordinates, so PQ = 8. The distance of T from $x - y - 1 = 0$ is $4/\sqrt{2} = 2\sqrt{2}$.

Step 4: Area $= \frac{1}{2}(8)(2\sqrt{2}) = 8\sqrt{2}$.

Common mistake: (A) halves the base or height again; (B) uses a horizontal or vertical separation as the perpendicular height; (C) omits the one-half in triangle area.

Q45

Mathematics · medium

Vectors / Dot/cross products and area of a parallelogram

Answer: (B) $3\sqrt{3}/2$ square units.

Given: $|\mathbf{a}| = |\mathbf{b}| = 1$ and the included angle is 60°.

Formula/Principle: Parallelogram area is the magnitude of the cross product of its adjacent sides.

Step 1: Expand $(\mathbf{a} + 2\mathbf{b}) \times (\mathbf{a} - \mathbf{b}) = -\mathbf{a} \times \mathbf{b} + 2\mathbf{b} \times \mathbf{a} = -3\mathbf{a} \times \mathbf{b}$.

Step 2: $|\mathbf{a} \times \mathbf{b}| = 1(1)\sin 60^\circ = \sqrt{3}/2$.

Step 3: The requested area is $3\sqrt{3}/2$.

Common mistake: (A) omits the linear-combination factor; (C) uses sin 60° = 1; (D) calculates triangle area instead of parallelogram area.

Physics

Method, reasoning and common errors

Q46

Physics · medium

Motion in a Straight Line / Distance and displacement from a signed velocity-time graph

Answer: (A) Average speed 2 m/s; average velocity zero.

Given: The piecewise linear graph crosses v = 0 at t = 3 s.

Formula/Principle: Displacement is signed area under v(t); distance is the area under |v(t)|.

Step 1: From 0 to 2 s, the triangular area is 4 m. From 2 to 3 s, it is another 2 m.

Step 2: The negative-velocity intervals 3 to 4 s and 4 to 6 s contribute signed areas -2 m and -4 m.

Step 3: Displacement is zero; distance is 4 + 2 + 2 + 4 = 12 m. Divide each by 6 s to get 0 and 2 m/s.

Common mistake: (B) interchanges speed and velocity; (C) misses area beside the central zero crossing; (D) treats negative velocity as positive when calculating displacement.

Q47

Physics · easy

Units and Measurements / Dimensional consistency of an empirical pressure model

Answer: (C) $ML^{-2}T^{-2}$.

Why: Pressure has dimensions force/area, or $ML^{-1}T^{-2}$. Every additive term must have pressure dimensions. Since h has dimensions L, $[b] = [P]/[h] = ML^{-2}T^{-2}$. The first term is also consistent because $[\rho v^2] = (ML^{-3})(L^2T^{-2})$.

Why not the others:

(A) Gives pressure dimensions without dividing by height.

(B) Gives acceleration dimensions, omitting density.

(D) Gives density dimensions, omitting acceleration.

Key point: Dimensional consistency applies separately to each additive term.

Q48

Physics · medium

Mechanical Properties of Fluids / Continuity and Bernoulli equation in a horizontal pipe

Answer: (B) $(15/2)\rho v^2$.

Given: The radius decreases by a factor of two, while elevation remains constant.

Formula/Principle: $A_1 v_1 = A_2 v_2$ and $P + \rho v^2/2$ is constant along a streamline.

Step 1: Area decreases by a factor of four, so the narrow-section speed is 4v.

Step 2: $P_1 - P_2 = \rho((4v)^2 - v^2)/2 = (15/2)\rho v^2$.

Common mistake: (A) doubles speed instead of quadrupling it; (C) omits the upstream kinetic-energy term; (D) mixes the area ratio with the kinetic-energy expression.

Q49

Physics · medium

System of Particles and Rotational Motion / Angular momentum conservation and actuator work

Answer: (D) $4\omega/3$ and $mR^2\omega^2/2$.

Given: No external axial torque, but the actuator may supply mechanical energy.

Formula/Principle: $I_i\omega = I_f\omega_f$; supplied work equals the change in total mechanical kinetic energy.

Step 1: $I_i = 3mR^2$ and $I_f = 2mR^2 + m(R/2)^2 = 9mR^2/4$.

Step 2: $\omega_f = (3/(9/4))\omega = 4\omega/3$.

Step 3: $K_i = 3mR^2\omega^2/2$, while $K_f = \frac{1}{2}(9mR^2/4)(4\omega/3)^2 = 2mR^2\omega^2$.

Step 4: Supplied work $K_f - K_i = mR^2\omega^2/2 > 0$.

Common mistake: (A) reverses the inertia ratio; (B) confuses conservation of angular momentum with conservation of kinetic energy; (C) assumes fixed angular speed without an external torque.

Q50

Physics · medium

Thermodynamics / Heat-engine energy balance and the Carnot bound

Answer: (A).

Why: The rejected heat is 800 - 300 = 500 J, satisfying the cyclic first-law balance. Actual efficiency is $300/800 = 0.375$. The maximum reversible efficiency is $1 - 300/500 = 0.40$. The given efficiency is below the bound, consistent with irreversibility. The reservoir entropy change is $500/300 - 800/500 = 1/15\,\mathrm{J\,K^{-1}} > 0$.

Why not the others:

(B) A reversible engine would attain 0.40 efficiency, not 0.375.

(C) Energy is conserved by the stated heat rejection.

(D) 0.375 is less than 0.40.

Key point: Test energy balance and the second-law limit separately.

Q51

Physics · medium

Oscillations / Spring force in vertical simple harmonic motion

Answer: (B) $3mg/2$.

Given: Equilibrium extension 0.10 m; initial displacement from equilibrium 0.03 m; initial speed divided by angular frequency 0.04 m.

Formula/Principle: SHM energy about equilibrium gives $A^2 = x_0^2 + (v_0/\omega)^2$, while $k(0.10) = mg$.

Step 1: $A = \sqrt{0.03^2 + 0.04^2} = 0.05$ m.

Step 2: The maximum extension from the natural length is $0.10 + 0.05 = 0.15$ m.

Step 3: The maximum spring tension is $k(0.15) = 3mg/2$. The minimum extension is 0.05 m, so the spring stays in tension.

Common mistake: (A) reports the maximum net restoring force about equilibrium; (C) adds displacement and velocity contributions linearly to overestimate the amplitude; (D) assumes equilibrium force balance holds throughout the motion.

Q52

Physics · medium

Waves / Resolving the ambiguity in a beat-frequency measurement

Answer: (C) 508 Hz and $(127/128)^2 T$.

Given: The initial string frequency is 512 Hz; the initial beat frequency is 4 Hz.

Formula/Principle: At fixed length and linear mass density, string frequency is proportional to the square root of tension. Beats equal the absolute frequency difference.

Step 1: The reduced string frequency is $512(255/256) = 510$ Hz.

Step 2: The first beat observation allows fork frequencies 508 or 516 Hz. Their differences from 510 Hz are 2 and 6 Hz, so only 508 fits both observations.

Step 3: For zero beats, the string must also be at 508 Hz: $T'/T = (508/512)^2 = (127/128)^2$.

Common mistake: (A) subtracts a second beat difference from the fork estimate; (B) reports the changed string frequency; (D) uses only the first beat measurement.

Q53

Physics · medium

Electric Charges and Fields / Work from a nonuniform electric-field graph and a turning point

Answer: (D) $7a/4$.

Given: The electric force opposes the initial motion throughout the shown positive-x region.

Formula/Principle: $K(x) - K(0) = q\int_0^x E_x(s)\,ds$.

Step 1: From 0 to a, the signed triangular field area is $-E_0 a/2$, so $K(a) = qE_0 a(5/4 - 1/2) = 3qE_0 a/4$.

Step 2: Beyond a, the field is constant at $-E_0$. Stopping requires $qE_0(x-a) = 3qE_0 a/4$.

Step 3: Thus $x = a + 3a/4 = 7a/4$, inside the indicated interval.

Common mistake: (A) uses the full initial energy with a constant field from x = 0; (B) miscounts the triangular area; (C) assumes the end of the drawn graph is a physical stopping boundary.

Q54

Physics · hard

Current Electricity / An unbalanced resistor bridge with an ideal ammeter

Answer: (B) $12/17$ A from A to B.

Given: The ideal ammeter makes A and B equipotential, but it can carry a nonzero current.

Formula/Principle: Apply Ohm's law and Kirchhoff's current law after accounting for the ammeter connection.

Step 1: Above the common middle node, $R_u = 2 \parallel 3 = 6/5\,\Omega$. Below it, $R_l = 4 \parallel 3 = 12/7\,\Omega$.

Step 2: The supply current is $12/(6/5 + 12/7) = 70/17$ A. The middle-node potential relative to the bottom is $(70/17)(12/7) = 120/17$ V.

Step 3: Current arriving at A through 2 ohm is $(12 - 120/17)/2 = 42/17$ A; current leaving A through 4 ohm is $30/17$ A.

Step 4: The excess $12/17$ A flows from A through the meter to B.

Common mistake: (A) uses the disconnected voltage-divider difference as a current; (C) reverses Kirchhoff's excess-current direction; (D) confuses zero meter voltage with zero meter current.

Q55

Physics · easy

Nuclei / Mass number and atomic number in alpha and beta-minus decay

Answer: (A) 234, 92.

Why: Alpha emission reduces mass number by four and atomic number by two, giving (234,90). Each beta-minus decay converts a neutron into a proton, increasing atomic number by one while preserving mass number. After two such decays the pair is (234,92).

Why not the others:

(B) Stops after the alpha decay.

(C) Accounts for beta-minus changes but omits alpha emission.

(D) Incorrectly decreases mass number by two rather than four in alpha decay.

Key point: Beta-minus emission changes the proton-neutron composition without changing the nucleon count.

Q56

Physics · medium

Electromagnetic Induction / Motional emf and terminal speed on inclined rails

Answer: (D) $mgR\sin\theta/(B^2l^2)$.

Given: Field is normal to the plane of motion and the total resistance is constant.

Formula/Principle: Motional emf $\mathcal{E}=Blv$; current $I=Blv/R$; magnetic braking force $F_B=BIl$.

Step 1: $F_B=B^2l^2v/R$, directed up the incline by Lenz's law.

Step 2: At terminal speed the net force along the rails is zero: $mg\sin\theta=B^2l^2v_t/R$.

Step 3: Solve to obtain the stated expression.

Common mistake: (A) uses the normal component of gravity; (B) inserts an inappropriate kinetic-energy factor of two into a force balance; (C) doubles the motional emf or gravitational component without justification.

Q57

Physics · medium

Ray Optics and Optical Instruments / Apparent depth through two plane liquid layers

Answer: (C) 7 cm; unchanged.

Given: Layer depths are 6 cm and 4 cm, with indices 3/2 and 4/3; the viewing medium is air.

Formula/Principle: In the paraxial approximation, apparent depth through parallel layers seen from air is $d_{\text{app}}=\sum_i t_i/n_i$.

Step 1: The contributions are $6/(3/2)=4$ cm and $4/(4/3)=3$ cm.

Step 2: Total apparent depth is 7 cm.

Step 3: Reversing layer order leaves the same two terms in the sum, so it does not change the result under the stated conditions.

Common mistake: (A) ignores refraction; (B) incorrectly treats interface order as changing the paraxial sum for fixed layer depths; (D) multiplies depths by indices instead of dividing.

Q58

Physics · medium

Alternating Current / Series RLC phasors and real power

Answer: (A) 10 V and 16 W.

Given: The reactive voltages are opposite in phase and the resistor voltage is in phase with current.

Formula/Principle: $V = \sqrt{V_R^2 + (V_L - V_C)^2}$; average power $P = IV_R$ for ideal L and C.

Step 1: Net reactive voltage is 10 - 4 = 6 V.

Step 2: Source voltage $= \sqrt{8^2 + 6^2} = 10$ V.

Step 3: Average power $= 2(8) = 16$ W; equivalently R = 4 ohm and $I^2R = 16$ W.

Common mistake: (B) adds all voltage magnitudes and equates apparent with real power; (C) uses $VI = 20$ VA as watts without a power factor; (D) combines the resistor and inductor voltages as in-phase scalars.

Q59

Physics · medium

Semiconductor Electronics: Materials, Devices and Simple Circuits / Half-wave rectification and mean power in a resistor

Answer: (B) $1/2$.

Given: The diode transmits one half-cycle unchanged and blocks the other.

Formula/Principle: Average resistor power is $\langle v^2 \rangle / R$, not $\langle v \rangle^2 / R$.

Step 1: For the direct sine wave, $\langle v^2 \rangle = V_0^2/2$, so $P_{\text{direct}} = V_0^2/(2R)$.

Step 2: Each half-cycle contributes equal squared-voltage area. Removing one gives $P_{\text{half}} = V_0^2/(4R)$.

Step 3: The power ratio is 1/2.

Common mistake: (A) reports an rms-voltage ratio rather than a power ratio; (C) halves an already averaged voltage and squares it; (D) ignores that half the cycle carries no current.

Q60

Physics · hard

Wave Optics / Two-colour interference, phase reversal and minimum slab thickness

Answer: (D) $3.0\,\mu\mathrm{m}$.

Given: Each colour begins with a relative phase of π; slab excess optical path is $\delta = (n-1)t = t/2$.

Formula/Principle: A bright fringe requires $\pi + 2\pi\delta/\lambda = 2\pi k$, so δ must be an odd multiple of $\lambda/2$ for each colour.

Step 1: We need $\delta = (2r+1)300\,\mathrm{nm} = (2s+1)500\,\mathrm{nm}$, with nonnegative integers r and s.

Step 2: The smallest common value with odd multipliers is $\delta = 1500$ nm: five times 300 and three times 500.

Step 3: $t = 1500/0.50 = 3000$ nm, or 3.0 micrometres.

Common mistake: (A) satisfies only the 600 nm condition; (B) uses an integer-wavelength condition appropriate to no initial phase reversal and still misses the second colour; (C) again satisfies only the 600 nm odd-half-wavelength condition. The two different colours need not interfere with one another.